# Identifying a Trial Population for Clinical Studies on Diabetes Drug Testing with Neural Networks

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**Abstract:** This project aims to model an end-to-end workflow of implementing different Artificial Intelligence (AI) tools for a clinical environment. A possible use case, such as the selection process of patients for a novel treatment, will be conducted by estimating the hospitalization time with a Neural Network on an Electronic Health Record (EHR) of diabetes. Then, Explainable AI (XAI) methods are computed for models trained with a Random Forest to evaluate the predictions. The diabetes readmission EHR dataset from the University of California, Irvine (UCI) Diabetes is used for this project. The trial population is selected by predicting the expected days for a person being hospitalized. An arbitrary boundary is set for choosing whether or not a patient shall be included into the trial. If so, a clear explanation of how the prediction is calculated and additional possible risk factors will be given in order to make the workflow explainable. This project shows that given a proper explanatory approach, AI can be a useful tool for the modern clinical environment. The workflow finally reveals that AI can be a beneficial support tool for doctors in the patient selection process.

**Keywords:** Machine Learning in the Industry 4.0; Clinical EDA; Data Analysis; AI in Medicine; Neural Networks; Diabetes

## 1 Introduction

Due to a new law in Germany, enforced on the first of January 2021 by the German Health Minister [Bu19], electronic patient files will be nationwide standardized in Germany. This opportunity can be seized to increase the impact of AI on the healthcare market to improve the health of Germany's population and facilitate doctor's work. Electronic Health Records offer a variety of different insights and application fields [SA19] [Am20], such as:

- Discover novel disease treatments.
- Improve patients diagnosis.
- Improve personalized healthcare.

In the year 2020, there are seven million cases of diabetes in Germany as stated by the Robert Koch Institut (RKI) [In20]. Even for well known diseases like diabetes, there is continuous research and novel drugs are invented on a frequent basis. Unfortunately, not every person is suitable for obtaining a treatment with a novel drug.

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The project's background is an artificially created scenario. The assumption is, that the database collected through the new law offers an easily accessible API to gather data from. Electronic Health Records are under high data privacy conditions. They can only be used if the data is completely anonymized and patients can not be traced back. It is further assumed that those issues have already been taken care of and therefore all limitations and regularity steps have been successfully established.

After preparing the EHR for this privacy requirements, the data can be utilized to gain insights and evaluations out of it. Using AI applications in hospitals is a highly classified realization, because wrong decisions can cause a dangerous outcome towards patients. For this reason, implementations of *AI in Medicine* require a very detailed explanatory analysis of the predicted outcome. Keeping this background in mind, the project is an implementation of an end-to-end workflow of how AI is used in the real clinical environment.

The first step is to figure out which patients are most likely suitable for receiving novel diabetes treatments. This can be achieved by a patient selection process, as stated by Dr. Toddenroth et. al. from the FAU [FK13] and a more recent paper from Szu Yeu Hu et. al. [SYH19]. Then an uncertainty estimation of the prediction is computed with the Aequitas toolkit from the University of Chicago. This toolkit is used to explain the predictions with Explainable Artificial Intelligence (XAI) methods. This process is supposed to enable an understanding of how certain an estimation is with respect to different dataset imbalance problems, e.g. demography. The research question for this project can therefore be concluded as:

*Can AI be safely applied in the real clinical environment if it provides enough explainability for its predictions?*

## 2 Methods

The dataset is originated from the UCI Machine Learning Repository [Re08] collected from 1999 to 2008 with over 100000 entries to train on. Various features such as demographics, diabetes conditions or medications are provided. In total, there are 55 features from which 36 are included into the modeling, because the other 19 features were not useable due to missing and NaN values. An important detail is the distribution of ethnicity, gender and age. This disparity is used in section *Results* for evaluating the uncertainty of the model. Figure 1 and 2 show, that the median age of diabetes patients is 70 to 80 years. Figure 1 shows that the gender can be ignored, because it is almost equally distributed among all patients. The ethnicity disparity reveals the biggest concern, because among different ethnicities there is a huge distribution imbalance. Asian, African and Hispanic people are very underrepresented in comparison to Caucasian people. This plays an important role for the later upcoming predictive analysis and explanatory analysis. Keeping these distributions

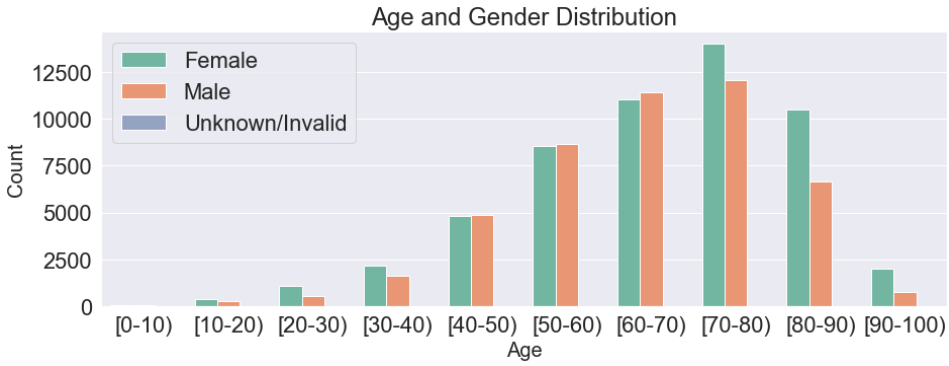


Fig. 1: The bivariate diagram shows the age disparity among all patients. The dataset is further split into the gender bins.

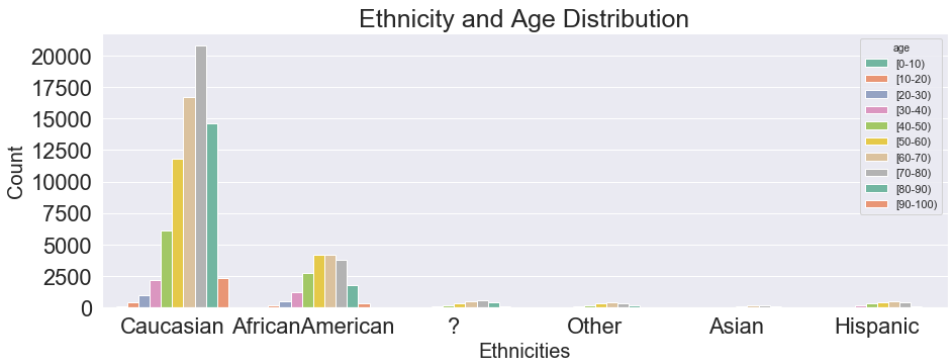


Fig. 2: The bivariate diagram shows the ethnicity disparity among all patients. The dataset is further split into the gender bins.

in mind, the methods used for this project will be conducted in the following respective order:

- Model a Neural Network and Random Forest.
- Explain the models with uncertainty estimation and metrics.
- Explain the predictions with SHAP and LIME.

SHapley Additive exPlanations (SHAP) is a python package for approximating the Shapely values. Those values can be used for various explanatory approaches, but for this project it is used for plots like in Figure 6. Local Interpretable Model-Agnostic Explanations (LIME) is a python package used for the LIME XAI approach.

Preprocessing is most of the time the first step before the training of an AI model. Various unnecessary features were dropped from the dataset and categorical features were transformed into a one-hot encoded representation. A supervised Neural Network always has a response variable, called a label. The *time in hospitalization* feature is used as the label, because it reveals the severity of each patient's diabetes disease. Usually, the longer people need to be hospitalized after receiving a treatment, the worse the condition of this person is. Predicting a low hospitalization time with a novel treatment can indicate compatibility with the patient and therefore he or she is a good match to be selected.

Neural Networks are able to achieve better results than standard Machine Learning approaches like the Random Forest. This can be proven due to the universal approximation theorem. Neural Networks are less explainable than most Machine Learning techniques, because they are mostly seen as a *blackbox*. Nevertheless, it was not feasible to compute the SHAP or LIME metrics for the Neural Network, that is why the Random Forest was used.

The Neural Network for this project is modeled with a Keras sequential function, consisting of Dense, Dropout, Dense Variational and a Distribution Lambda layers to estimate the hospitalization time with a certain probability. The Distribution Lambda layer from Tensorflow incorporates an ensemble of models which reports the variability of the prediction. It can be viewed as taking the mean of various regressions and measures then the qualitative difference between all outputs. Those layers are usually used in the clinical environment to reduce the risk of:

- Aleatoric Uncertainty: statisical uncertainty (*known unknowns*).
- Epistemic Uncertainty systematic uncertainty (*unknown unknowns*).

Aleatoric uncertainty will be evaluated with respect to the ethnicity distribution of the data. The neural networks prediction will be measured based on different evaluation metrics:

- Precision, Recall and F1 Score.

- Brier Score.
- AUC Score.

The loss function for the Neural Network is the negativ log likelihood function [DZ18]:

$$\text{Loss} = -log(y) \text{ for each prediction } y$$

The loss generally increases when the regression is unclear about its prediction and shrinks with increasing certainty. At this point, the model is able to predict how long a person is going to stay in the hospital with respect to the demographics and other features.

In conclusion to guarantee full transparency of the patient selection process, Shapely values and LIME try to explain which features led to which output. In other words, which features have a positive and a negative impact on the hospitalization time. The key difference between these two XAI methods is that Shapely values want to explain *how* a feature contributes to the overall prediction, while leaving out other combinations of features. LIME wants to explain what the most important features were in general for the prediction.

## 3 Results

After having the workflow in mind, the first step is to train the Neural Network and evaluate its results. The model trained for 50 epochs and reached a loss of 2.97. Figure 3 shows the course of events for different evaluation metrics. For example, the *Brier Score* itself is best when it is as low as possible. The other scores are best when they are as close to 100

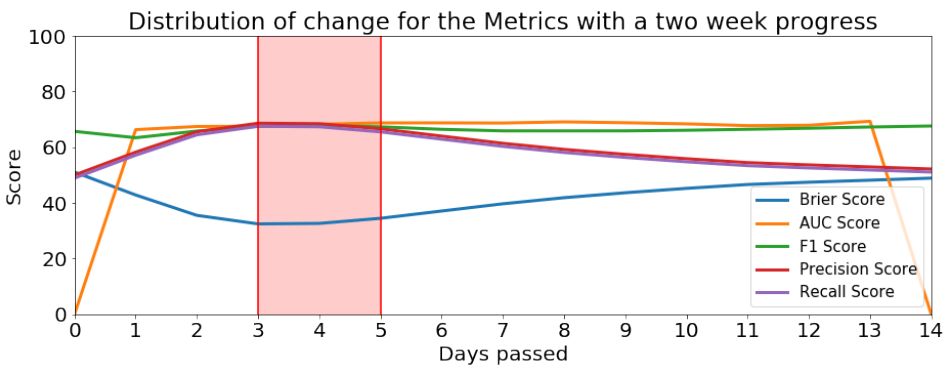


Fig. 3: The graphic shows the temporal change of the metrics with respect to a different time constraints on the boundary. The best values for the metrics can be found between the third and fifth day. Setting the boundary somewhere in the red highlighted area produces the best and most certain estimation.

percent as possible. The x-Axis represents the predicted time of hospitalizations in days. All metrics have an optimal score for the boundary at day four, as can be seen in Figure 3. After selecting the most promising discrimination boundary at day four, the uncertainty estimation with respect to ethnicity can be computed.

Figures 4 and 5 highlight that the true positive rate for Caucasian people is higher compared to the other ethnicities. The false-negative rate of other ethnicities is higher compared to the reference Caucasian people.

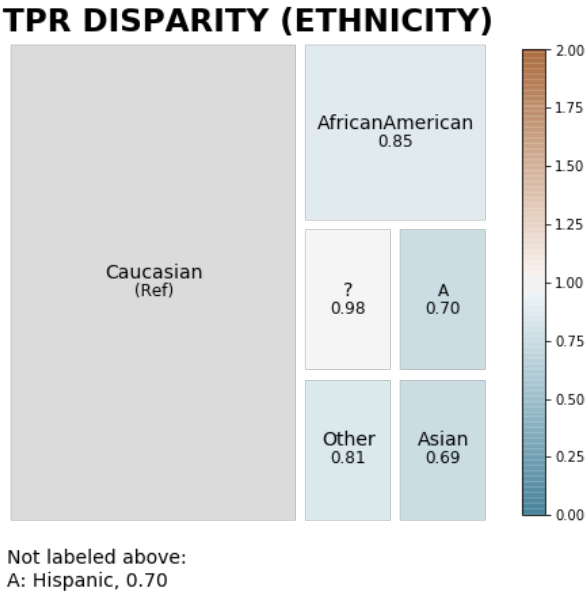


Fig. 4: True positive rate among ethnicities with Caucasians as reference group with the discrimination boundary at day four.

The next step is the investigation of how each prediction is computed and what the most important features for those predictions are. SHAP and LIME are used for this purpose, as mentioned in the Section Methods. Unfortunately, it was not possible to compute these two XAI methods with the Neural Network model, because the computational time due to many parameters was not feasible within the time constraint. To still generate valueable insights, a Random Forest was used to train on the same train- and testdata like the Neural Network. The Random Forest has fewer parameters and therefore the computation is much faster. The scores for the prediction of the hospitalization time were slightly worse.

The second interpretational approach are the Shapely values, which can be seen in Figure 6. This Figure lists 20 features with both their positive and negative impact on the prediction of the hospitalization time. Negative (blue) values on the x-Axis have an decreasing impact

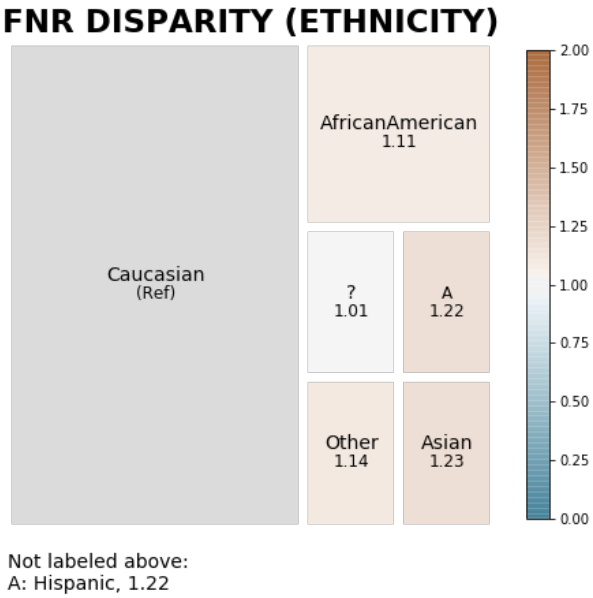


Fig. 5: False negative rate among ethnicities with Caucasians as reference group with the discrimination boundary at day four.

on the hospitalization time and positive (red) values on the x-Axis will increase the time of hospitalizations.

## 4 Discussion

Training the model alone is not sufficient enough for the selection of a patient into the trial population. There needs to be a specific boundary indicating after which time of hospitalization the diabetes is so serious, that this person can be taken into consideration for the trial. If this boundary is not set, the selection process can not be accomplished. Figure 3 not only shows the course of events, but it also highlights that different boundaries for the x-Axis produce various score curves. The optima is between the third and firth day.

The uncertainty estimation can be evaluated for different demographics and boundaries against each other. Figures 4 and 5 indicate, that if a patient is being selected and he or she is Caucasian, the probability of a positive or negative choice is more accurate compared to other ethnicities. Looking back on Figure 2, this important prediction disparity can be explained by the huge imbalance of ethnicities in the training data.

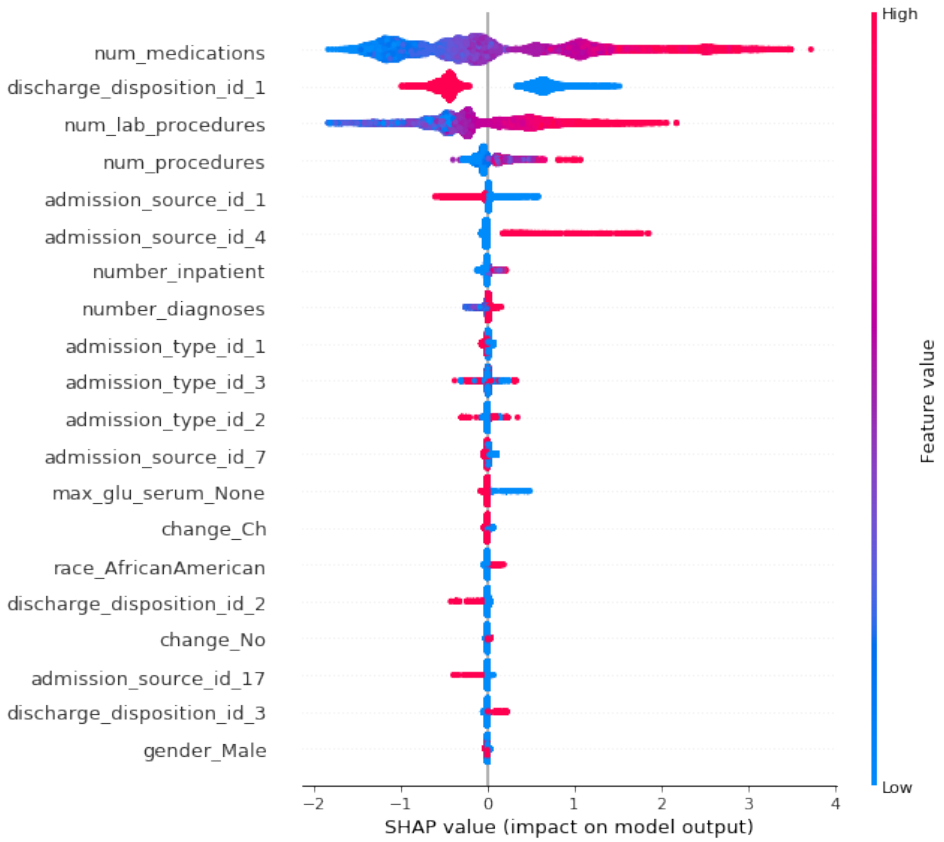


Fig. 6: Important features computed by SHAP.

Shapely values and LIME values need to be interpreted in a different way than the metrics. Figure 7 illustrates the single components which led to the prediction of 6.42 days. The meaning of these categorical features are provided by the following bullet points:

- discharge_disposition_id_1: Discharged to home.
- admission_source_id_5: Transfer from a Skilled Nursing Facility (SNF).
- admission_source_id_4: Transfer from a hospital.
- admission_source_id_10: Transfer from critial access hospital.

This patients EHR data has the features *admission_type_id_1* and discharge_disposition_id_6. This means that **not** having the features admission_source_id_5 and 6 actually has a **positive** (fewer days in hospital) impact on the hospitalization time. In contrary, when a patient has the feature *discharge_disposition_id_1*, he or she usually stays longer in the hospital. Also e.g. if the patient is taking the medication *Migitol*, he or she generally will stay up to one day (0.87) longer in hospital than patients who do not take this drug. Explanations like this can be conducted for all features to further investigate new insights into which effects medications and treatments can have on the patients health. This can be highly informative knowledge when making the choice to select patients into a clinical trial.

The distribution of *num_medications* can be seen in Figure 8. This Figure shows a right skewed normal distribution. Based on this distribution, it can be seen that a normally distributed numerical feature can have both a negative and a positive impact on the prediction of the most important features computed by SHAP, as shown in Figure 6. The more medications a person takes, the worse the condition and the longer the hospital stay for this person is, and vice versa. Whereas the feature *admission_source_id_4* always has a positive impact on the hospitalization time (higher hospitalization time).

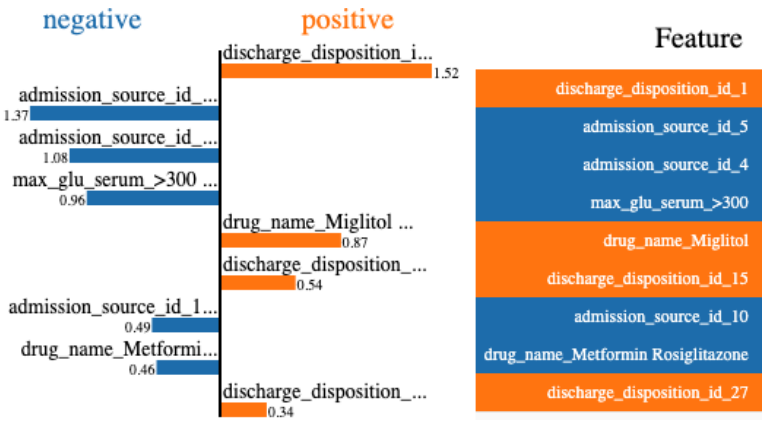


Fig. 7: Important features computed by LIME.

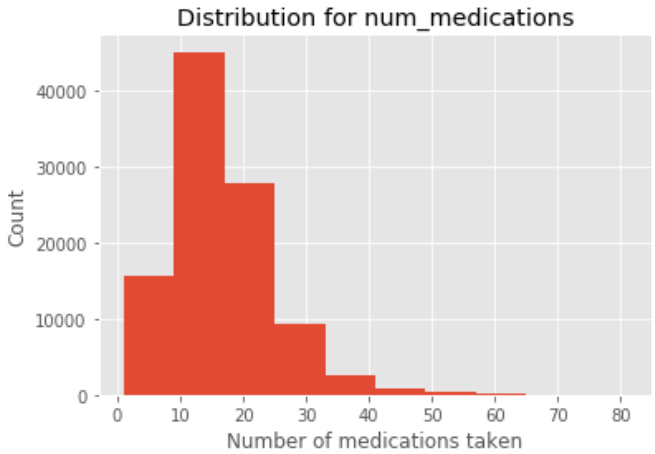


Fig. 8: Normal right skewed distribution of the feature *num_medications*.

After taking all evaluations and explanatory steps into consideration: Can AI safely be applied in the Industry 4.0 and especially hospitals? Keeping the results in mind, it appears that AI shows many valueable traits.

**Pros**:

- Support for doctor's decision.
- Patient preselection saves a lot of time.
- New insights into effects of medications or other features on patients.

**Cons**:

- Unstructured EHR data is very hard to handle.
- Not sure if every EHR dataset can be used to gain information out of it.
- The minimum necessary needed data amount for highly explainable predictions is unclear.

These techniques can be a good decision-support technology for doctors. If there are for example 2000 patients stored in the hospital's database, this preselection process can shrink the number of possible suitable patients already down in advance. Doctors will make the final decision based on their experience, but they can end up saving a lot of time by looking at already e.g. 20 preselected patients instead of investigating 2000 patient files. So even if the predicted population is not going to be selected, it has no dangerous impact for patients

in general. Doctors should be taught to develop more trust into a well trained and good performing AI systems, because they could benefit from it.

## 5 Conclusion

The conclusion of this entire workflow leads to three final questions to be answered.

*Can it be applied without additional doctor's approval?*

At this day and age not. What if only one person dies because of a false decision of the AI's prediction? This is a very sensitive topic, that's why the law and regulatory clinical instances like the FDA in the United States or the Arzneimittelbehörde in Germany must be involved when implementing such technologies in the clinical environment.

*Can it be applied as a support technology?*

If a doctor validates the results, it can be used to support the doctors decision for their clinical trials and save them up a lot of time.

*Does a combination of metrics improve the overall explainability and how many metrics are enough for a sufficient explainable approach?*

Using at first the F1-Score for evaluating the model's performance and then computing the uncertainty estimation with respect to critical features shows reasonable model decision making. Furthermore it can be seen, that many different approaches reveal different insights into the prediction. A good combination of metrics can enhance the overall performance, but there is no gold standard of metrics that explain 100 percent of the models behavior.

## 6 Future Work

For future research further improvements can also be made, such as:

- Try different XAI methods or explain more features in different ways.
- Try different Neural Network architectures with GPU training for higher accuracies.
- Model also the Epistemic Uncertainty with different Neural Network Layers: For example: Tensorflow offers the DenseVariational layer for a more robust Network prediction, as explained in Section Methods (epistemic uncertainty).
- Apply SHAP and LIME to a Neural Network.

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